

NGHIA H. NGUYEN

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GitHub $\diamond$ LinkedIn

RESEARCH INTERESTS

Machine Learning, Optimization, Interpretability, Representation Geometry and Algebraic Structure in Foundation Models

EDUCATION

University of Pennsylvania, Philadelphia, PA *Aug. 2023 – Present*

PhD in Computer Science, advised by René Vidal

MS in Computer and Information Science *Aug. 2023 – Aug. 2026*

University of Science, VNU-HCM, HCM City, Vietnam

Bachelor of Science in Computer Science *Sep. 2017 – Nov. 2021*

SELECTED WORKS

(* indicates co-first authors.)

- “On the Provable Multi-Modal Transfer of Hierarchical Concept Structure in CLIP Embeddings,” **Nghia Nguyen**, Tianjiao Ding, Lachlan Ewen MacDonald, René Vidal, *NeurIPS*, 2026. [paper link]
- “Hierarchical Concept Embedding & Pursuit for Interpretable Image Classification,” **Nghia Nguyen**, Tianjiao Ding, René Vidal, *CVPR*, 2026; **Spotlight** at the TRUE-V Workshop. [paper link] [code]
- “SSD: Sparse Semantic Defense against Semantic Adversarial Attacks to Image Classifiers,” **Nghia Nguyen**, Darshan Thaker, Konstantinos Emmanouilidis, Tianjiao Ding, René Vidal, *UCRL Workshop @ ICLR*, 2026.
- “Improving Neural Ordinary Differential Equations with Nesterov’s Accelerated Gradient Method,” **Nghia H. Nguyen***, Tan M. Nguyen*, Huyen K. Vo, Stanley Osher, Thieu N. Vo, *NeurIPS*, 2022. [paper link] [code]
- “Towards a Robust WiFi-based Fall Detection with Adversarial Data Augmentation,” Tuan-Duy H. Nguyen, **Huu-Nghia H. Nguyen**, *CISS*, 2020.

RESEARCH EXPERIENCE

FPT Software AI Center, Ho Chi Minh City, Vietnam *Oct. 2021 – Jul. 2023*
AI Research Resident (Predoctoral Researcher)

- Advisors: Tan M. Nguyen (UCLA, currently NUS), Thieu N. Vo (Ton Duc Thang University, currently University of Bath).
- Derived the continuous adjoint equation for the Nesterov Neural ODE, an important theoretical result for our work.
- Implemented the method and ran experiments of Nesterov Neural ODE on image classification, continuous normalizing flows, and point cloud separation.

VinBrain, Ho Chi Minh City, Vietnam
Applied Scientist Intern

Apr. 2021 – Aug. 2021

- Supervisor: Soan T.M. Duong (University of Wollongong).

- Improved the object detection model mAP@0.5 score by over two times from a pretrained model when tested on a custom version of the Open Images v6 dataset.

LSIR, École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland (Remote from Vietnam) *May 2020 – Oct. 2020*
Research Intern

- Supervisor: Chi Thang Duong (EPFL).
- Worked on a Subtree-Reuse-based SQL Query Optimizer using Graph Neural Network and Box Embeddings.
- Performed experiments on the Attribute Discovery dataset to verify the plausibility of a new Multimodal Feature Fusion method using Graph Neural Networks, achieving a test accuracy of 96.86% on this dataset.

Faculty of Information Technology, University of Science, VNU-HCM, Ho Chi Minh City, Vietnam *Sep. 2019 – Aug. 2020*
Research Assistant

- Discovered the correlation between forking patterns, software health, and risks in software repositories.
- Performed inference in the Software Heritage Dataset to extract forking patterns between 50,000 repositories using PostgreSQL and Neo4J.

ACHIEVEMENTS & AWARDS

- Scholar Award; *Advances in Neural Information Processing Systems (NeurIPS)* 2022. (Granted full registration fee and accommodation in New Orleans.)
- Fellow; The Vietnam Education Foundation (VEF) 2.0 Program 2022. (Awarded to excellent Vietnamese PhD applicants after two interview rounds with leading Vietnamese academics and US professors.)
- Participant; Pre-PhD Summer School by VinUni-Illinois Smart Health Center, VinUniversity 2022 (Top 50/220 applicants).
- Dean's List; Advanced Program in Computer Science, University of Science, VNU-HCM 2018, 2020 (Top 5/92 students).
- Honorable Mention; Vietnam National Olympiad for Undergraduates in Informatics 2019.
- 2nd place; University of Science VNU-HCM Olympiad in Informatics 2019.

OTHER EXPERIENCE

Big-O Coding, Ho Chi Minh City, Vietnam *Jul. 2018 – Dec. 2020*
Teaching Assistant, Problem Setter, Software Engineer

- Explained Algorithms and Data Structure concepts to students weekly and prepared over 20 programming problems.
- Built version 1 of the Front-end Web Application for Big-O Coding's website to host programming problems and contests for up to 1,900 users using ReactJS.
- Reduced page traversing paths for administrators by 50% by implementing a shortcut to problems.

Pathfinder Vietnam, Ho Chi Minh City, Vietnam *Sep. 2018 – May 2020*
Founder, Product Team Leader, Software Engineer

- Led product team of 6 in implementing the Minimum Viable Product using ReactJS and NodeJS.

- Introduced the product's core feature by implementing the programming testing system.

SKILLS

- Programming: Python, C++, JavaScript, Java (prior experience).
- Web: HTML, CSS, ReactJS, Redux.
- Machine Learning: TensorFlow, Neural ODEs, adversarial robustness.
- Databases & Tools: SQL, PostgreSQL, Neo4J.
- English: IELTS Academic 8.5/9.0 (Reading: 9.0, Listening: 9.0, Writing: 7.5, Speaking: 7.5).

SIDE PROJECTS

- **What The Food app** (2018): Implemented a Convolutional Neural Network to identify 101 types of food using TensorFlow; converted the Keras model to a TensorFlow Lite model and deployed it on an Android application using Firebase ML Kit.

EXTRACURRICULAR ACTIVITIES

- Reviewer: NeurIPS, ICLR, CVPR, ICML, ECCV.
- Volunteer: ICLR 2020, ICML 2021.